Tanay Shah

Robotics & Control Engineer

RESUME_1871

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Total Experience: 5 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 5 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: ROS 2, Python, Motion Planning, Robot Kinematics, Embedded Systems, Simulink, SLAM, OpenCV, PLC, C++, MATLAB, PID Control

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Senior Robotics & Control Engineer | Urbanaut Engineering | Jan 2023 - Present

- Improved pick-position repeatability by 15% after calibration updates
- Validated 171 robot simulation scenarios with recorded sensor logs
- Lowered navigation recovery events by 23% using improved obstacle handling
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Robotics & Control Engineer | KoraLink Innovations | Jul 2021 - Dec 2022

- Lowered navigation recovery events by 18% using improved obstacle handling
- Reduced robotic path completion time by 16% through route and controller tuning

PROJECTS

Autonomous Mobile Robot | Robot Kinematics, PID Control, Embedded Systems, ROS 2

- Built ROS 2 nodes for localization, obstacle avoidance and route execution.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Warehouse Mapping Trial | SLAM, Simulink, Robot Kinematics, ROS 2

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Western Plains Institute of Technology, 2021

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Industrial Robot Safety (2025)